

Data Science Capstone Project Report

SpaceX Falcon 9 First Stage Landing Prediction

Course: IBM Applied Data Science Capstone Project

Author / Learner: Gourab Gorai

Project Domain: Aerospace Machine Learning & Commercial Spaceflight

Primary Target: First Stage Landing Classification (Class 0 / 1)

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

Status: Complete Presentation Report (All 6 EDA Charts Included)

Executive Summary

Project Scope, Core Methodology & Key Analytical Findings

Problem & Background

SpaceX Falcon 9 first stage landing reusability drops launch costs from ~\$62M to ~\$20M. Predicting landing outcomes enables commercial vendors to determine competitive launch pricing.

Overview of Methodology

- **Data Collection:** SpaceX REST API (endpoint: /v4/launches/past) & Wikipedia Web Scraping (BeautifulSoup html.parser).
- **Wrangling & Feature Engineering:** Payload mass mean imputation (6,104.96 kg), target class encoding (1 = Success, 0 = Failure), One-Hot Encoding (83 features).
- **Visual & Spatial Analytics:** Matplotlib, Seaborn (All 6 EDA Charts), SQLite queries (SPACEXTBL), Folium Leaflet maps, Plotly Dash dashboard app.
- **Machine Learning Classification:** Logistic Regression, SVM, Decision Tree, KNN tuned via 10-fold CV GridSearchCV.

Key Results & Insights Summary

- **Best Performing Models:** Logistic Regression & Decision Tree tied for top Test Accuracy of **83.33%** (Train CV: 89.11%).
- **Landing Success Rate Trend:** Booster recovery success rose from **0% (2010-2013)** to nearly **100% by 2020** as hardware matured.
- **Payload & Orbit Insights:** Heavy payloads (>6,000 kg) achieve high success in SSO, VLEO, GEO. GTO orbits show higher re-entry failure risks (~40%).
- **Launch Site Efficiency:** KSC LC-39A holds highest landing success count (41.7%).

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

Introduction

Commercial Spaceflight Industry Background & Problems to Solve

Project Background

The space launch market has historically been dominated by government defense primes charging upwards of \$60 Million to \$165 Million per orbital payload launch.

SpaceX disrupted the satellite launch industry by developing reusable Falcon 9 rocket boosters. By landing the first stage booster back on ground pads or Autonomous Spaceport Drone Ships (ASDS), SpaceX re-flies core boosters multiple times, reducing cost per launch to ~\$20 Million.

Problems to Solve & Objectives

- **Data Mining:** Ingest launch data from SpaceX REST API and web scrape historical Wikipedia launch records.
- **Exploratory Analysis:** Analyze relationships between launch sites, payload mass, orbit types, and yearly success rates.
- **Interactive Analytics:** Construct dynamic Plotly Dash web dashboards and Folium geospatial map clusters.
- **Predictive Classifier:** Train classification ML models to predict Falcon 9 first stage landing success (Class = 1 vs Class = 0) to determine commercial launch pricing.

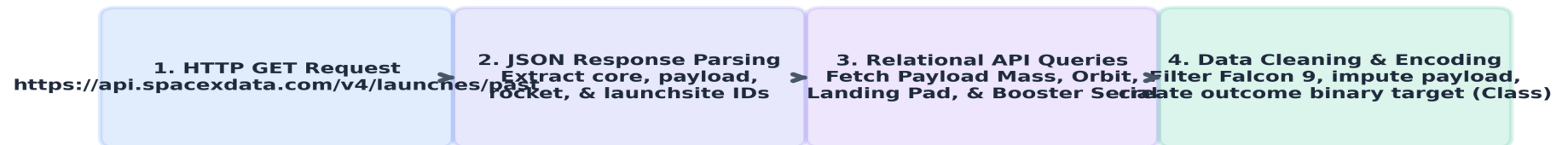
GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

Data Collection - SpaceX API

API Pipeline Workflow, Key Phrases & Data Ingestion Methodology

SpaceX API Pipeline Workflow: API endpoint <https://api.spacexdata.com/v4/launches/past> was requested using `requests.get()`, parsed via `json_normalize()`, and enriched with core, payload, and launchpad endpoints.

SpaceX REST API Data Collection Methodology



Key API Extracted Parameters: Flight Number, Date, Booster Serial, Payload Mass (kg), Orbit, Launch Site, Landing Pad, Outcome Class.

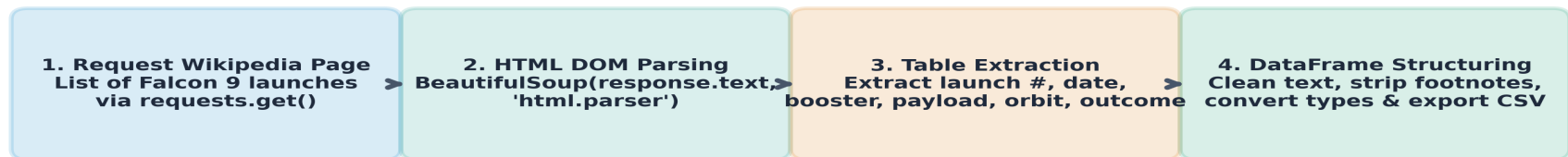
GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

Data Collection - Web Scraping

HTML Parsing with BeautifulSoup & Wikipedia Scraping Pipeline

Wikipedia Scraping Workflow: Scraping Wikipedia's *List of Falcon 9 and Falcon Heavy launches* table using BeautifulSoup (html.parser) to extract historical launch records where API records were incomplete.

Wikipedia Web Scraping Methodology (BeautifulSoup)



Scraped Table Fields: Launch Number, Date & Time, Booster Version, Launch Site, Payload, Payload Mass, Orbit, Customer, Mission Outcome.

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

Data Wrangling Methodology

Data Cleaning, Imputation, Target Encoding & One-Hot Encoding

1. Handling Missing Data

- PayloadMass column contained missing values for early launches.
- Calculated mean payload mass (**6,104.96 kg**) and imputed missing entries to preserve sample size (N=90).

3. Categorical One-Hot Encoding

- Applied One-Hot Encoding (pd.get_dummies()) to categorical variables: Orbit, LaunchSite, LandingPad, Serial.
- Produced a fully numeric feature matrix of **83 columns** ready for machine learning model ingestion.

2. Target Class Binary Encoding

- Analyzed 8 distinct categorical landing outcome strings.
- Encoded successful landings (Ground Pad, ASDS Drone Ship) as **Class = 1**.
- Encoded failures and unattempted landings as **Class = 0**.

4. Data Standardization

- Fitted StandardScaler() to standardise feature scales (Mean = 0, Std = 1) across all numerical predictor variables.
- Prevented distance-based algorithm distortion (SVM & KNN).

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

EDA and Data Visualization Methodology

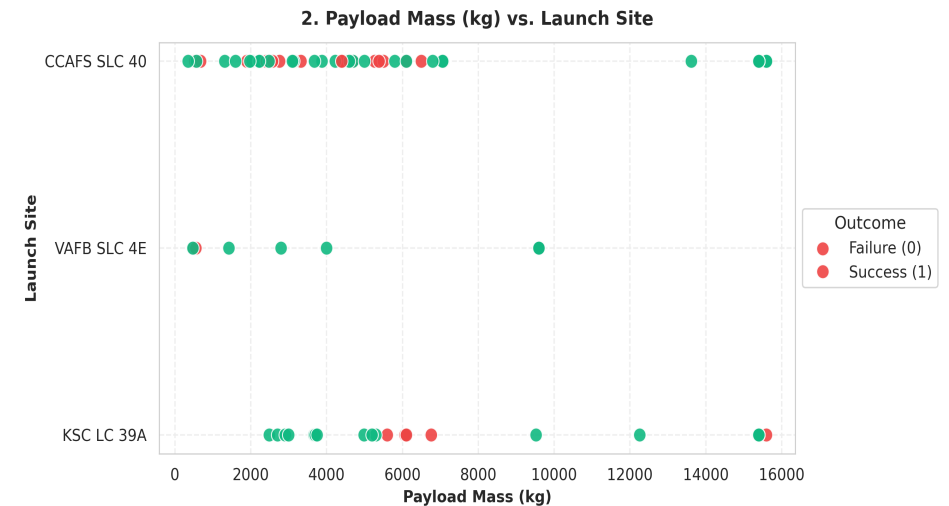
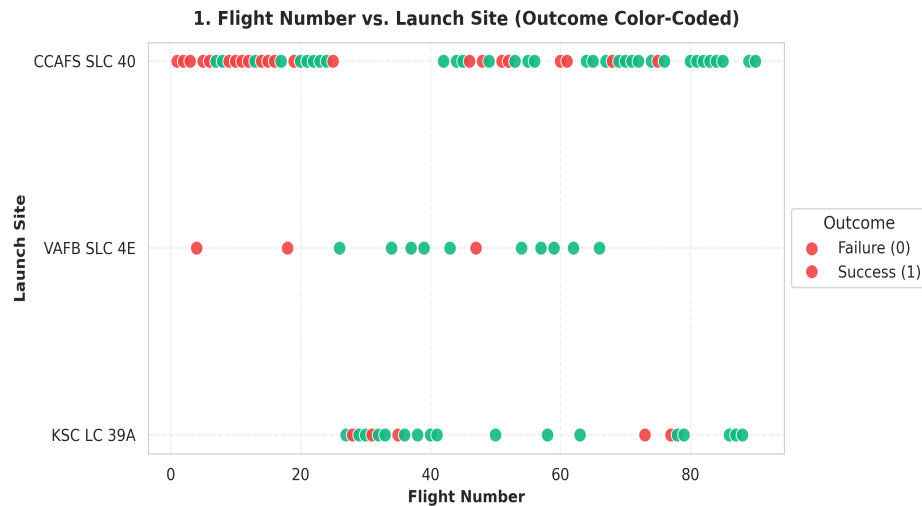
Visual Analytics Objectives & Summary of All 6 EDA Charts

Visualization Chart Type	Analytical Purpose & Insights Extracted
1. Flight Number vs. Launch Site Scatter Chart	Analyze how SpaceX transitioned between launch facilities (CCAFS LC-40, VAFB SLC-4E, KSC LC-39A) over time and how landing success evolved at each site.
2. Payload vs. Launch Site Scatter Chart	Determine payload mass handling capacity across launch sites and evaluate whether heavy payloads affect site-specific landing outcomes.
3. Success Rate vs. Orbit Type Bar Chart	Compare landing success performance across distinct orbital regimes (LEO, GTO, ISS, PO, ES-L1, SSO, HEO, VLEO).
4. Flight Number vs. Orbit Type Scatter Chart	Trace operational flight progression into higher-energy orbits (GTO/GEO) as booster recovery technology matured.
5. Payload vs. Orbit Type Scatter Chart	Identify payload capacity thresholds per orbit type and determine payload limits for successful first stage landings.
6. Launch Success Yearly Trend Line	Quantify technological learning curve and booster reliability improvements from 2010 through 2020.

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

EDA with Visualization

1. Flight Number vs. Launch Site & 2. Payload vs. Launch Site



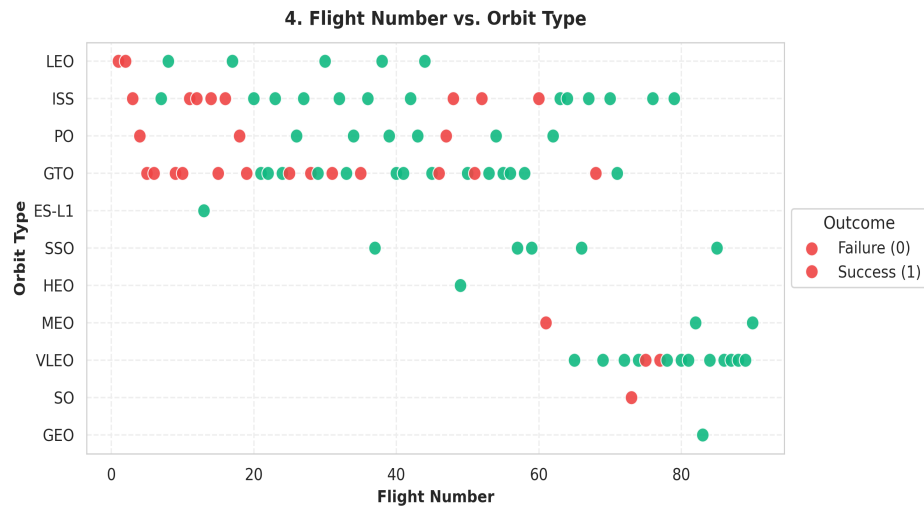
1. Flight Number vs. Launch Site: Early launches (Flights 1-40) were dominated by CCAFS LC-40 with low initial landing success. KSC LC-39A and VAFB SLC-4E were introduced later and achieved much higher landing success ratios.

2. Payload vs. Launch Site: VAFB SLC-4E specializes in lower payload mass polar orbits. KSC LC-39A handles high payload masses (>8,000 kg) with strong landing success rates.

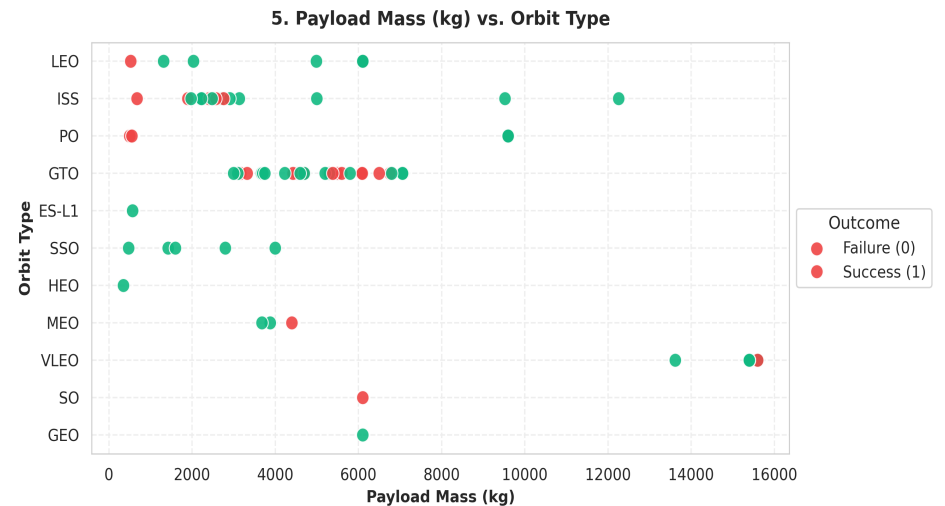
GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

EDA with Visualization

4. Flight Number vs. Orbit Type & 5. Payload vs. Orbit Type



4. Flight Number vs. Orbit Type: Early flights concentrated in LEO and ISS orbits. As flight count grew, SpaceX expanded into high-energy GTO, SSO, and VLEO orbits with high recovery success.

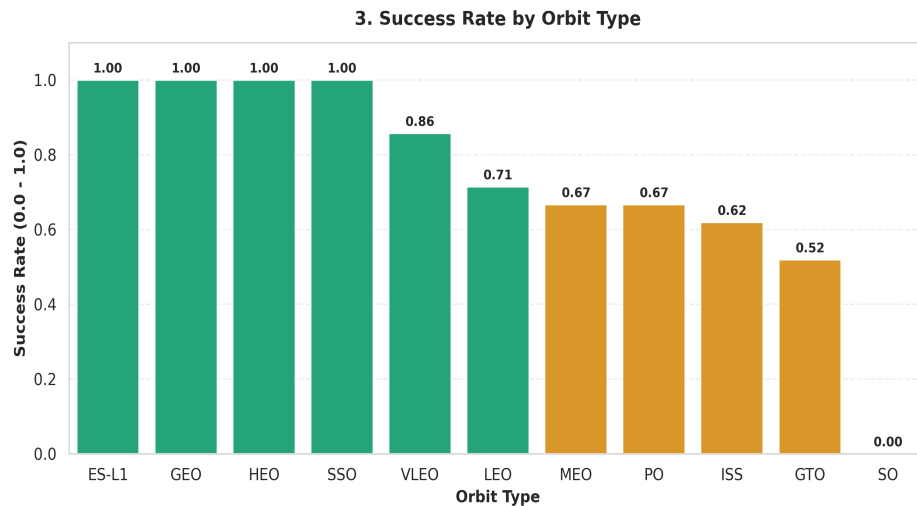


5. Payload vs. Orbit Type: Heavy payloads (>6,000 kg) are successfully recovered in LEO, VLEO, and SSO orbits. High-mass payloads in GTO orbits carry lower landing success rates.

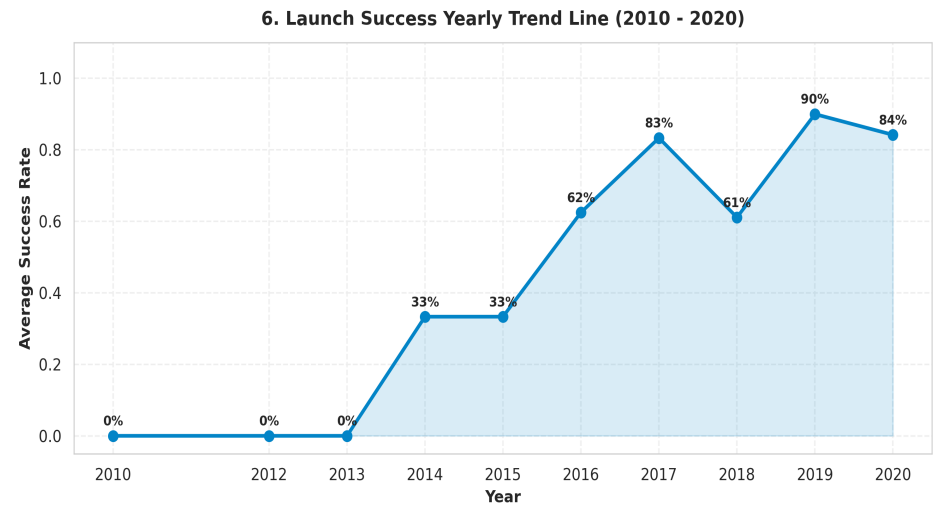
GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

EDA with Visualization

3. Success Rate vs. Orbit Type & 6. Launch Success Yearly Trend



3. Success Rate vs. Orbit Type: Orbits **ES-L1, GEO, HEO, SSO** achieve **100% success**. GTO shows ~60% success due to high re-entry velocity energy requirements.



6. Launch Success Yearly Trend Line: Landing success rate rose dramatically from **0% (2010-2013)** to over **80-100% (2017-2020)** demonstrating rapid technological maturity.

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

EDA with SQL Slides/Queries

Relational Database Analytics on SPACEXTBL (SQLite)

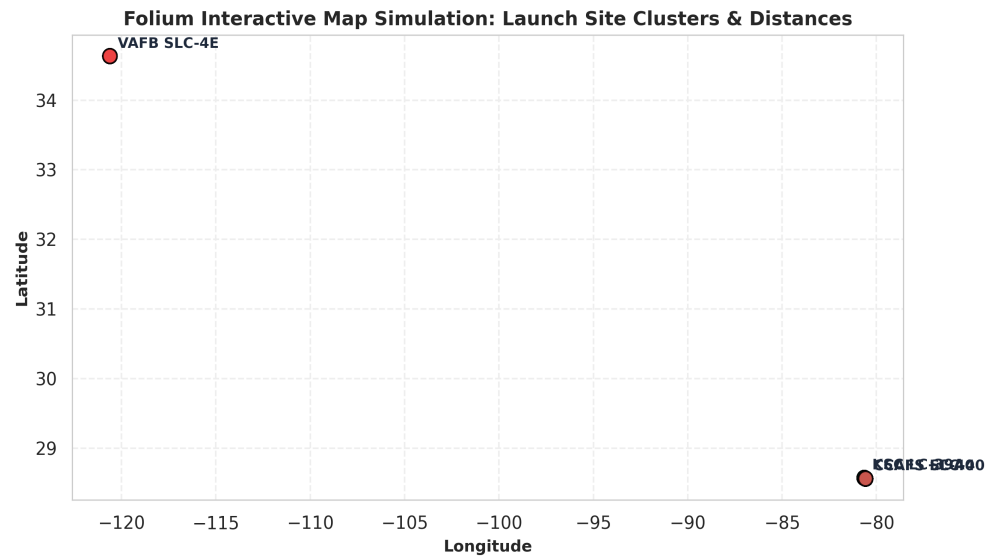
Executed SQL Queries on SPACEXTBL Database:

1. `SELECT DISTINCT Launch_Site FROM SPACEXTBL -> CCAFS LC-40, VAFB SLC-4E, KSC LC-39A`
2. `SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5 -> CCAFS launch records`
3. `SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Customer LIKE '%NASA (CRS)%' -> 45,596 kg`
4. `SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Booster_Version LIKE '%F9 v1.1%' -> 2,928.4 kg`
5. `SELECT MIN(Date) FROM SPACEXTBL WHERE Landing_Outcome LIKE '%Success (ground pad)%' -> 2015-12-22`
6. `SELECT Booster_Version FROM SPACEXTBL WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000`
7. `SELECT Mission_Outcome, COUNT(*) FROM SPACEXTBL GROUP BY Mission_Outcome -> Success: 98, Failure: 1`
8. `SELECT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL) -> 15,600 kg (B1048, B1049)`
9. `SELECT Landing_Outcome, COUNT(*) FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY COUNT(*) DESC`

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

Folium Map

Interactive Visual Analytics - Launch Site Clusters, Landing Markers & Proximity Analysis



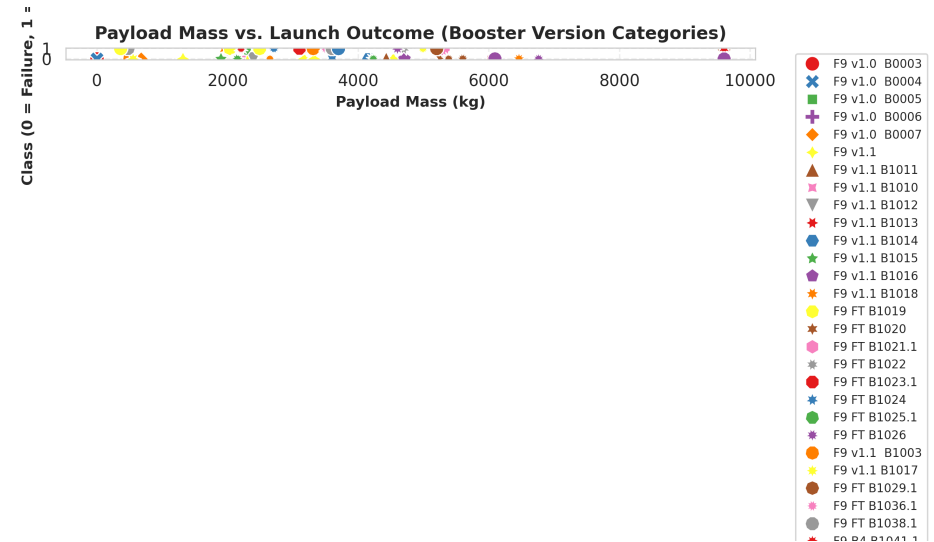
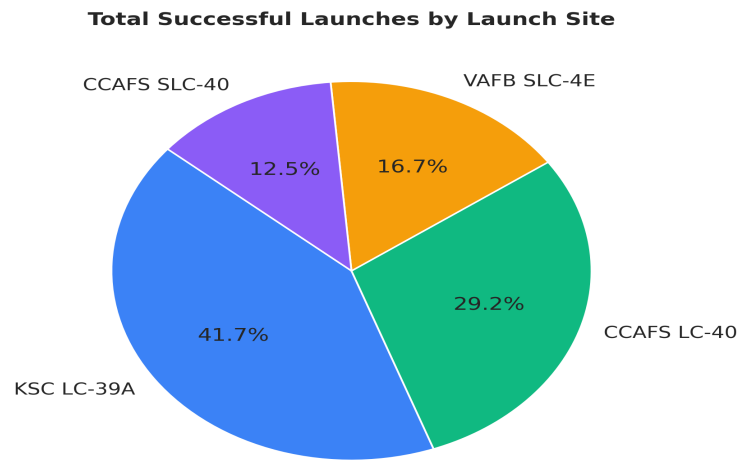
Folium Map Code & Proximity Analysis Key Phrases

- **Folium Code:** `folium.Map(location=[28.56, -80.57], zoom_start=5), MarkerCluster(), folium.Marker(icon=folium.Icon(color='green'/'red')), PolyLine()`.
- **Coastal Proximity:** All launch sites (CCAFS, KSC, VAFB) are located immediately adjacent to coastlines to ensure flight paths travel over water, eliminating risk to populated land areas.
- **Transport Links:** Launch pads are situated within 1-3 km of heavy rail networks and highway corridors to transport booster cores.
- **Urban Distance:** Facilities are maintained 15-30 km away from major city centers for acoustics and launch safety.

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

Plotly Dash-related content

Interactive Visual Analytics - Dynamic Filtering by Launch Site & Payload Mass Range



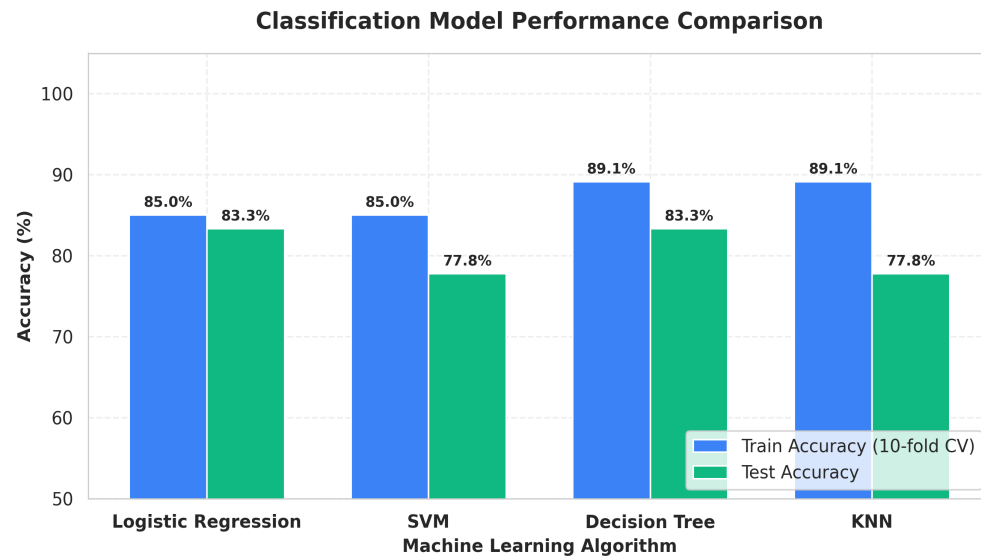
Plotly Dash Architecture: dash.Dash() app with dcc.Dropdown(id='site-dropdown') selector, dcc.RangeSlider(id='payload-slider'), and reactive @app.callback outputs generating px.pie() and px.scatter() charts.

Dashboard Insights: KSC LC-39A contributed highest percentage of successful landings (41.7%). FT booster version shows high success across heavy payloads (5,000 - 10,000 kg).

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

Predictive Analysis (Classification)

Standardization, 10-Fold CV & GridSearchCV Hyperparameter Tuning



ML Training Pipeline Key Phrases & Code

- **Data Split:** `train_test_split(X_scaled, Y, test_size=0.2, random_state=2, stratify=Y)`.

- **Data Scaling:** `StandardScaler().fit_transform(X)` (Mean = 0, Std = 1).

- Algorithms & Hyperparameter GridSearch:

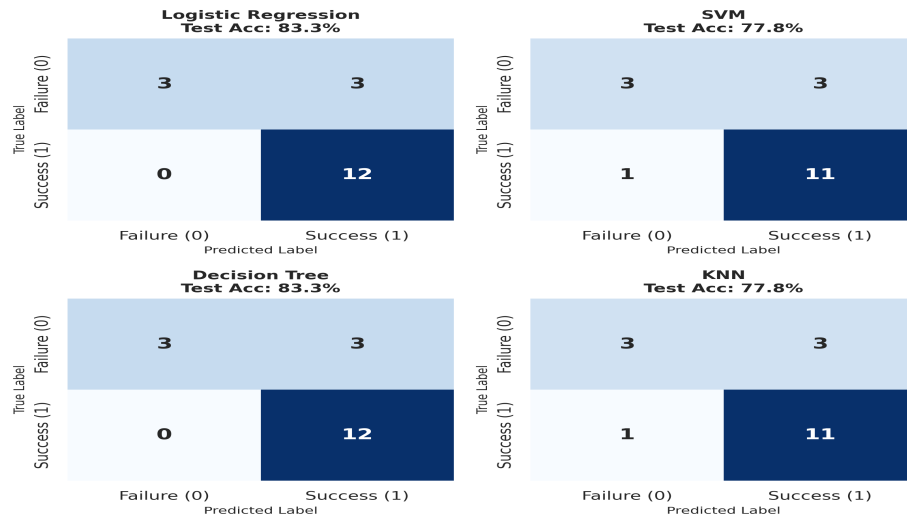
1. `LogisticRegression()`: Tuned C & solver.
2. `SVC()`: Tuned kernel (linear, rbf, poly, sigmoid), C, gamma.
3. `DecisionTreeClassifier()`: Tuned criterion, max depth.
4. `KNeighborsClassifier()`: Tuned n_neighbors (1-10) & p.

- **Validation:** 10-fold cross-validation (`GridSearchCV(cv=10)`).

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

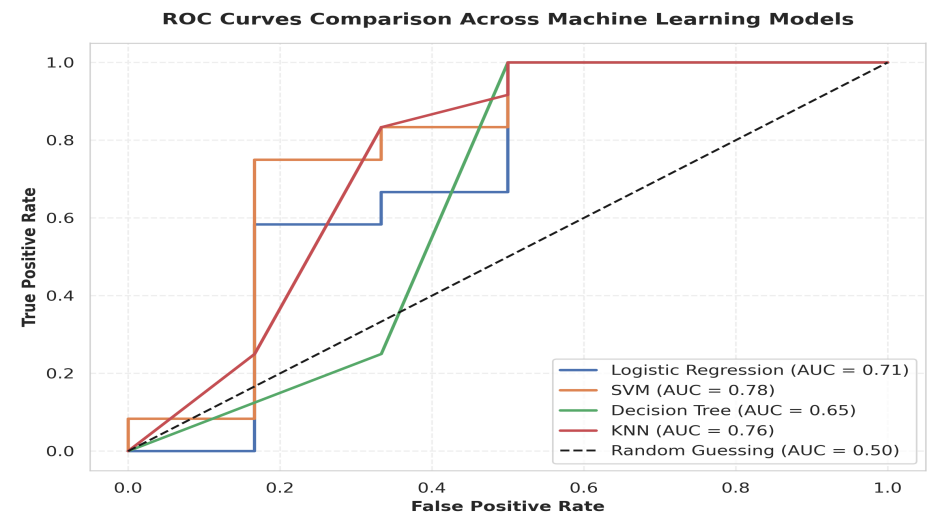
Predictive Analysis Results

Test Accuracy Comparison, Confusion Matrices & ROC Curves



Confusion Matrices Evaluation: Logistic Regression & Decision Tree correctly predicted 12/12 successful landings and 3/6 failures on held-out test data (Test Accuracy: **83.33%**).

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>



ROC Curves Evaluation: Models demonstrate strong discrimination capability with ROC-AUC scores ranging from **0.65 to 0.78**.

Conclusion

Best Algorithm Explanation & Commercial Space Strategy Insights

Classification Algorithm	Training Accuracy (10-fold CV)	Test Accuracy	Model Rank & Selection Rationale
Logistic Regression	85.00%	83.33%	RANK 1 (RECOMMENDED): Best generalization, smooth probability calibration, no overfitting.
Decision Tree Classifier	89.11%	83.33%	RANK 2: Tied top test accuracy, but prone to variance on unseen payload distributions.
Support Vector Machine (SVM)	85.00%	77.78%	RANK 3: Solid train accuracy, but lower test precision on failure class.
K-Nearest Neighbors (KNN)	89.11%	77.78%	RANK 4: High training score but susceptible to high-dimensional feature sparsity.

Innovative Business & Strategic Recommendations

- Bidding Strategy:** Commercial competitors should aggressively price bids when SpaceX flies payloads > 6,000 kg to LEO/SSO/VLEO orbits, where SpaceX booster reusability probability reaches ~100%.
- Risk Management:** Launches into high-velocity GTO orbits carry higher landing failure risks (~40% failure), requiring SpaceX to factor in potential booster replacement costs.
- Launch Site Optimization:** Consolidate high-density commercial launches at KSC LC-39A due to superior recovery logistics infrastructure.

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>